**PRODUCT REQUIREMENTS DOCUMENT (PRD)**
**Product Name:** Hashmar CropEx Digital Farmers' Identity Platform
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## 1. Product Overview

The Hashmar CropEx Digital Farmers' Identity Platform is a multi-module digital ecosystem designed to register, verify, and empower **smallholder farmers** through identity, financial inclusion, farm data, and market access.
It transforms farmers from *invisible participants* into *bankable, traceable economic actors*.

## 2. Problem Statement

Smallholder farmers in Nigeria face systemic barriers:

- ☐ No verifiable identity → cannot access loans
- ☐ No structured farm data → high risk for investors
- ☐ Limited access to markets → low income
- ☐ No insurance or pension → high vulnerability

Result?
**A massive agricultural engine running without oil.**

## 3. Product Goals & Objectives

**Primary Goal**
Digitally onboard **500–1000 farmers within 6 months**.

**Key Objectives**

- ☐ Create verified digital farmer identities
- ☐ Enable access to financing and credit scoring
- ☐ Provide farm-level data for decision-making
- ☐ Improve farmer income and productivity
- ☐ Connect farmers to structured markets

## 4. 👥 Target Users

**Primary Users**

- Smallholder farmers (low digital literacy in some cases)

**Secondary Users**

- Financial institutions (banks, MFIs)
- Agribusinesses / offtakers
- Government agencies
- Insurance & pension providers

## 5. Product Features (Modules)

**1. Farmer Identity Module**

- ☐ Unique Farmer ID
- ☐ Biometric registration
- ☐ Personal + cooperative data

Think of this as "BVN for farmers"

**2. Farm Mapping Module**

- GPS mapping
- Farm size + soil data
- Satellite verification

Turns farmland into *data-rich assets*

**3. Credit & Financing Module**

- Credit scoring engine
- Loan access (inputs, equipment)
- Financial partner integrations

Converts trust into numbers banks can act on

**4. Extension & Training Module**

- WhatsApp advisory
- Video training
- AI support assistant
- Field officer integration

A farmer's pocket agronomist

**5. Health Insurance Module**

- HMO integrations
- Low-cost insurance enrollment

**6. Pension Module**

- Micro pension contributions
- Seasonal payment structure

**7. Market Linkage Module**

- Offtaker connections
- Aggregation logistics
- Price visibility

Where effort finally meets reward

**6. User Flow**

1. Farmer is onboarded via field agent
2. Biometric + personal data captured
3. Profile is created (ID issued)
4. Farm is mapped (GPS + satellite)
5. Farmer gains access to:
    - Loans
    - Training
    - Insurance
    - Market access

**7. Product Requirements**

**Platforms**

- Mobile App (Field Agents)
- Web Dashboard (Admin & Partners)
- SMS/WhatsApp Interface (Farmers)

**Core Functional Requirements**

- Offline data capture (critical for rural areas)
- Secure biometric authentication
- Real-time data sync when online
- Credit scoring algorithm
- Partner API integrations
- Low bandwidth optimisation

**Non-Functional Requirements**

- High data security (farmer identity + financial data)
- Scalable architecture (target: 100k farmers)
- Multi-language support (local languages later phase)

**8. Success Metrics (KPIs)**

- Number of farmers registered
- Number of farms mapped
- % of farmers accessing loans
- Insurance enrollment rate
- Pension adoption rate
- Increase in farmer income

Feature 1

**1. Farmer Identity Module**

- [ ] Unique Farmer ID Biometric registration
- [ ] Personal + cooperative data

# MODULE 1: FARMER IDENTITY

*Product Requirements Document*

## 1. Module Overview

Create a unique, biometrically-verified digital identity for every onboarded farmer that is portable, tamper-proof, and interoperable across all Hashmar CropEx platform modules and external partners.

## 3.2 Key Objectives

- [ ] Register 500–1,000 verified farmer identities within the first 6 months
- [ ] Achieve 100% biometric uniqueness … zero duplicate farmer IDs
- [ ] Capture rich personal, cooperative, and land-related data at enrollment
- [ ] Enable offline enrollment to support low-connectivity rural areas
- [ ] Generate a scannable, portable Farmer ID card (physical + digital)
- [ ] Serve as the master identity layer that all other modules reference

## 4. Target Users

| User Type | Role | Interaction with Module |
|---|---|---|
| Smallholder Farmer | Primary | Gets enrolled, receives Farmer ID, updates personal info |
| Field Agent / Enumerator | Primary | Conducts biometric capture, completes enrollment forms |
| Cooperative Officer | Secondary | Links farmers to cooperative, manages group data |
| Platform Admin | Secondary | Reviews, approves, or flags farmer profiles |
| Financial Institution | Secondary | Queries verified identity for loan eligibility checks |

| | | |
|---|---|---|
| Government Agency | Secondary | Accesses anonymised data for policy, subsidy targeting |
| Insurance Provider | Secondary | Validates farmer identity for policy enrollment |

## 5. Functional Requirements

### 5.1 Unique Farmer ID

Each registered farmer receives a system-generated, non-repeatable Farmer ID upon successful completion of biometric and personal data capture.

| Attribute | Specification |
|---|---|
| ID Format | HCX-[STATE_CODE]-[YYYY]-[6-DIGIT RANDOM] e.g. HCX-LG-2025-004821 |
| Uniqueness | System enforces no-duplicate constraint at generation time |
| Immutability | ID never changes once assigned; farmer data can be updated separately |
| Portability | ID is embedded in QR code on physical card and digital wallet |
| Cross-module | All platform modules (Credit, Insurance, Market) reference this ID as FK |
| Offline Assign | Pre-assigned ID batches pushed to field agent devices for offline use |

### 5.2 Biometric Capture

Biometric data serves as the primary deduplication and authentication mechanism, ensuring one farmer = one ID.

| Biometric Type | Capture Method | Use Case |
|---|---|---|
| Fingerprint (primary) | 10-finger capture via portable scanner | Identity authentication & deduplication |
| Facial Recognition | Front-facing photo via mobile app camera | Profile photo + liveness check |
| Fallback | 2-finger capture + photo if full scan fails | Rural areas with degraded fingerprints |

Biometric rules:

- Biometric data is encrypted at rest (AES-256) and in transit (TLS 1.3)
- No raw biometric images stored — only encrypted templates
- NDPR-compliant storage with explicit farmer consent captured digitally
- Deduplication check runs automatically against existing farmer DB at sync

### 5.3 Personal Information needed

Field agents capture structured personal data via the mobile app enrollment form.

| Field | Type | Required? |
|---|---|---|
| Full Legal Name | Text | Mandatory |

| | | |
|---|---|---|
| Date of Birth | Date picker | Mandatory |
| Gender | Dropdown (M/F/Other) | Mandatory |
| Phone Number | Numeric + validation | Mandatory |
| State of Origin | Dropdown (36 states + FCT) | Mandatory |
| Local Government Area | Cascading dropdown | Mandatory |
| Residential Address | Free text | Mandatory |
| NIN (National ID No.) | Numeric, 11 digits | Mandatory (BVN as fallback) |
| BVN | Numeric, 11 digits | Mandatory |
| Marital Status | Dropdown | Mandatory |
| Education Level | Dropdown | Mandatory |
| Years of Farming Experience | Numeric | Mandatory |
| Primary Crop(s) | Multi-select | Mandatory |
| Secondary Crop(s) | Multi-select | Mandatory |
| Next of Kin Name | Text | Mandatory |
| Next of Kin Phone | Numeric | Mandatory |
| Next of Kin Relationship | Dropdown | Mandatory |
| Profile Photo | Camera capture | Mandatory |

**5.4 Cooperative & Association Data**

Farmers may belong to one or more cooperatives. This data links individual identities to collective economic units.

| **Field** | **Type** | **Required?** |
|---|---|---|
| Cooperative Name | Text / searchable dropdown | Mandatory |
| Cooperative Registration No. | Alphanumeric | Mandatory |
| Membership Role | Dropdown (Member / Secretary / Chairman / Treasurer) | Mandatory |
| Date Joined | Date picker | Mandatory |
| LGA of Cooperative | Dropdown | Mandatory |
| Commodity Focus | Multi-select | Mandatory |
| Cooperative Size (members) | Numeric | Mandatory |
| Land Ownership Type | Dropdown (Owned / Leased / Communal / Family) | Mandatory |
| Approximate Farm Size (hectares) | Numeric | Optional (pre-mapped in Farm Module) |
| Input Supplier Name | Text | Optional |

### 5.5 Digital & Physical ID Card

- Digital card available immediately post-enrollment in-app and as PDF
- Physical card printed on request (PVC, 85.6 x 54mm standard)
- Card contains: Name, Photo, Farmer ID, QR Code, Cooperative, Crop type, Expiry date
- QR code links to verified farmer profile (read-only public view for partners)
- Card expires every 2 years; renewal triggers re-verification

## 6. Non-Functional Requirements

| Category | Requirement |
|---|---|
| Security | AES-256 encryption at rest; TLS 1.3 in transit; biometric templates never stored raw |
| Privacy (NDPR) | Explicit consent captured; data subject rights (access, deletion) supported |
| Offline Support | Full enrollment possible with no internet; syncs automatically when online |

## 7. User Flows

| Step | Actor | Action |
|---|---|---|
| 1 | Field Agent | Opens mobile app and selects 'New Farmer Enrollment' |
| 2 | Field Agent | Captures biometrics: fingerprint scan (10 fingers) + facial photo |
| 3 | System | Checks biometrics against local cache for duplicates |
| 4 | Field Agent | Completes personal data form (name, DOB, NIN, address, etc.) |
| 5 | Field Agent | Optionally completes cooperative data section |
| 6 | Field Agent | Reviews summary screen and farmer provides verbal consent |
| 7 | System | Generates Farmer ID (or uses pre-assigned offline ID) |
| 8 | System | Stores record locally and queues for server sync |
| 9 | Farmer | Receives digital ID card via SMS link or field agent shows on screen |
| 10 | System | On connectivity: syncs to server, runs full dedup, activates profile |
| 11 | Admin | Reviews flagged profiles (if any) and approves or requests re-enrollment |

### 7.2 Farmer Profile Verification Flow (Partner / Lender)

- Partner calls Hashmar CropEx API with Farmer ID or scans QR code
- System returns verified profile: name, photo, ID status, cooperative, primary crop, farm size
- Sensitive fields (NIN, BVN, biometrics) never exposed via partner API
- Each API call is logged with requester identity and timestamp

### 7.3 Farmer Profile Update Flow

- Field agent logs into app and searches farmer by ID or name
- Updates permitted: phone number, address, cooperative, crop type, next of kin
- Immutable fields: Farmer ID, date of birth, biometrics (requires admin override + re-enrollment)
- All updates require field agent authentication and create an audit log entry
  - Farmers should be able to self onboard

## 10. Success Metrics (KPIs)

| KPI | Target (6 months) | Measurement Method |
|---|---|---|
| Farmers enrolled & verified | 500–1,000 | Admin dashboard enrollment count |
| Duplicate registration rate | < 0.5% | Biometric dedup alerts in system log |
| Enrollment time per farmer | < 5 minutes | App-measured session duration |
| Offline enrollment success rate | 100% | Sync success logs |
| NIN verification success rate | > 85% | NIMC API response logs |
| Farmer ID cards issued | = enrollments | Card generation count |
| Partner API queries on farmer ID | > 50/month (M3+) | API access log |
| Field agent adoption rate | > 90% | Agents using app vs enrolled |

### 11.1 Assumptions

- Field agents will have Android smartphones (OS 8.0+) with functional cameras
- Portable fingerprint scanners are available to all field agents at launch
- Farmers have or can obtain NIN before or during enrollment
- At least intermittent internet connectivity available once per day per agent for sync
- NIMC API access can be obtained via accreditation or partner agreement

### 11.2 Constraints

- NDPR compliance is non-negotiable — no biometric data exposed to third parties
- Platform must function in areas with 2G/3G connectivity or no connectivity at all
- Initial build targets Android only; iOS and web admin in Phase 2
- Budget and timeline: MVP enrollment module within 3 months of kick-off

## 12. Out of Scope (v1.0)

- Credit scoring algorithm (Module 3)
  - GPS farm boundary mapping (Module 2)
  - Farmer-facing self-service app (farmers cannot self-register; field agent required in v1)
  - Multi-language UI — English only in v1 (Hausa, Yoruba, Igbo in v2)

- Iris scan biometrics — fingerprint + facial only in v1
- Physical ID card printing infrastructure — digital card only in v1

**13. Risks & Mitigations**

| Risk | Severity | Mitigation |
|---|---|---|
| Degraded fingerprints (elderly/manual labour farmers) | High | 2-finger fallback + facial recognition as primary for affected farmers |
| NIMC API downtime or rejection of accreditation | High | Proceed with offline enrollment; NIN verification done in async batch |
| Field agents submitting fraudulent enrollments | High | Geo-tagging, photo capture of farmer with agent, admin review queue |
| Low farmer trust / resistance to biometric capture | Medium | Community sensitisation, use of local language, cooperative officer as proxy |
| Data breach of farmer PII | High | AES-256 encryption, zero raw biometrics stored, strict role-based access |
| Duplicate IDs due to sync conflicts | Medium | Conflict resolution logic with timestamp priority; admin review for duplicates |

**Appendix A: Farmer ID Format Specification**

**Format: HCX - [STATE_CODE] - [YEAR] - [6-DIGIT SEQUENCE]**

- HCX — Hashmar CropEx platform prefix
- STATE_CODE — 2-letter abbreviation (e.g., LG = Lagos, KN = Kano, AB = Abia)
- YEAR — 4-digit enrollment year (e.g., 2025)
- 6-DIGIT SEQUENCE — zero-padded random sequence with system-enforced uniqueness

Example IDs:

- HCX-LG-2025-004821 — Lagos farmer, enrolled 2025, sequence 4821
- HCX-KN-2025-000001 — Kano farmer, first enrolled in that state-year

**Appendix B: Consent Statement (Draft)**

The following consent statement will be read to the farmer by the field agent and acknowledged verbally and via digital signature/thumbprint on the enrolment device:

> *"I, [Farmer Name], consent to the collection, storage, and use of my personal information and biometric data by Hashmar CropEx Limited for the purpose of creating and maintaining my digital farmer identity profile. I understand that my data will be used to facilitate access to financial services, agricultural support, and market linkage. I understand I may request access to or deletion of my data at any time by contacting Hashmar CropEx Limited. I confirm that this information has been explained to me in a language I understand."*

*Note: This is a draft. Legal review required before use.**